

AATIF ALI KHAN

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Islamabad, Pakistan

Objective

Cybersecurity student (6th Semester) with practical experience in security engineering, threat detection, and malware analysis, developing systems such as SOAR platforms, DLP engines, and forensic tools. Seeking a cybersecurity internship to apply hands-on technical skills in building and securing scalable, real-world systems.

Projects

• Automated Threat Detection & Response System (SOAR)

Nov 2025

Suricata, pfSense, n8n, Docker, WireGuard

- Engineered a hybrid-cloud SOAR platform integrating Suricata IDS and pfSense, automating threat responses to enforce **internal security policies**.
- Developed a risk scoring algorithm mapping network alerts to **compliance frameworks**, reducing false positives and maintaining an accurate incident log for audit evidence.
- Automated the generation of human-readable incident reports, streamlining the **internal audit** process and ensuring transparent compliance tracking.

• MFT Recovery Tool: Advanced Digital Forensics Utility

Apr 2026

Python, PySide6, Raw Disk I/O, Data Carving, NTFS Parsing

- **Engineered** a digital forensics desktop application in Python, leveraging **raw physical disk I/O** to bypass OS-level logical corruption and execute **forensically sound data extraction** from **unallocated space**.
- **Architected** a custom Master File Table (MFT) parsing and signature carving engine, performing **sequential byte-level sweeps** to identify, validate, and recover **permanently deleted NTFS records**.
- **Developed** a highly responsive GUI utilizing PySide6, integrating a **real-time hexadecimal viewer** and an **automated directory tree reconstruction** algorithm to significantly accelerate incident response workflows.

• Distributed AI Meeting Automation System

May 2026

Python, Selenium, Faster-Whisper, Kaggle GPUs, Ngrok, SciPy

- Engineered a distributed, autonomous meeting agent using **Python** and **Selenium**, implementing advanced DOM manipulation and anti-bot evasion techniques to autonomously navigate and authenticate into Google Meet and Microsoft Teams.
- Architected a low-latency, cloud-offloaded **Speech-to-Text (STT)** pipeline by capturing and resampling continuous audio streams via **SciPy**, transmitting data to a remote **Kaggle GPU** server via an **ngrok** tunnel to eliminate local compute bottlenecks.
- Deployed the **Faster-Whisper** model for real-time inference and developed an event-driven keyword detection algorithm that parses live transcriptions, autonomously triggering asynchronous **ActionChains** to inject human-like chat responses.

• DocuMind AI: Event-Driven Multi-Doc RAG System

Jan 2026

FastAPI, Inngest, Qdrant, Streamlit, OpenAI API, Docker, uv

- Architected a scalable, event-driven **RAG** backend utilizing **FastAPI** and **Inngest** to orchestrate durable, retrievable background workflows for high-throughput PDF ingestion.
- Integrated **Qdrant Vector DB** and **OpenAI's text-embedding-3-large** model to ensure high-precision semantic retrieval and zero-hallucination grounding, optimizing deployment with **Docker** and **uv**.
- Developed a unique **Multi-Document Comparison Intelligence** mode that leverages metadata-filtered semantic search to perform side-by-side mathematical analysis of financial data across disparate sources.

• Hashed Based Data Loss Prevention (DLP) Engine

May 2026

Python, Flask, SQLite, SHA3-256, SimHash Cryptography

- Engineered a zero-knowledge Data Loss Prevention (DLP) platform utilizing **SHA3-256 exact matching** and **SimHash fuzzy logic** to autonomously detect and block unauthorized exfiltration of sensitive corporate assets.
- Architected a highly secure, role-based backend implementing a dual-database schema to strictly isolate administrative credentials from immutable forensic audit logs, aligning with **governance and compliance standards**.
- Designed an advanced Context-Triggered Piecewise Hashing (CTPH) algorithm capable of identifying obfuscated data leaks with a Hamming Distance threshold, significantly mitigating **insider threats**.

• SecureShare: Secure File-Sharing Platform

April, 2026

Node.js, Express.js, MySQL, JWT, VirusTotal API, Cryptography



- Engineered a secure full-stack file-sharing application utilizing Node.js and MySQL, implementing Envelope Encryption (KMS simulation) to ensure robust data confidentiality at rest.
- Integrated the VirusTotal API and custom Steganography Detection algorithms (Chi-Square, RS Analysis) to autonomously scan and block malicious file payloads in real-time.
- Architected a highly secure backend featuring Role-Based Access Control (RBAC), Email-based MFA, and robust input validation to defend against SQL Injection, XSS, and brute-force attacks.

• Automated ML for Deep Learning-based Malware Detection

May 2026

Python, PyTorch, TensorFlow, DARTS, CNNs, FFNNs, PyQt5, EMBER-2018



- Developed an end-to-end malware analysis platform integrating a **Static FFNN detector** (trained on 800,000 EMBER-2018 samples) and a **Dynamic DARTS-inspired CNN monitor** for live runtime process behavior.
- Implemented a Neural Architecture Search (NAS) and Hyperparameter Optimization (HPO) pipeline for the static detector, extracting 1,649 PE features to calculate malware probabilities and threat risk scores with **96.2% accuracy**.
- Engineered a custom data pipeline transforming live process metrics (CPU, Memory, I/O, Network) into $1 \times 64 \times 64$ behavior tensors; optimized the architecture using Differentiable Architecture Search (DARTS), achieving a **0.993 AUC** and **95.0% accuracy**.
- Built a cybersecurity-themed command center GUI using **PyQt5** for real-time risk visualization, live event logging, and automated HTML/JSON forensics report generation.

• Tour & Travel Booking Platform (MERN Stack)

Dec 2025

MongoDB, Express.js, React.js, Node.js, JWT, Nodemailer



- Engineered a full-stack **MERN** booking application following MVC architecture, utilizing **React Context API** to efficiently manage complex global state for user profiles and live bookings.
- Implemented secure backend authentication and authorization pipelines using **JSON Web Tokens (JWT)** and Bcrypt, ensuring robust data integrity and secure user sessions.
- Integrated **Nodemailer** with Gmail SMTP to build an automated, event-driven notification system for instant booking confirmations and user onboarding.

• Pac-Man Assembly Edition

Dec 2024

Tools: Assembly (Irvine32)



- Developed a classic Pac-Man game from scratch in Assembly, implementing real-time collision detection and smooth gameplay mechanics.
- Programmed ghost AI using the Manhattan Distance algorithm for intelligent movement.

Skills

- **Programming Languages:** C++, Python, JavaScript (ES6+), Assembly (x86/MASM), Bash Scripting
- **Security Operations & Tools:** Suricata IDS, pfSense Firewall, n8n (SOAR), WireGuard VPN, Burp Suite, Wireshark, Nmap, Procmon, Procdot, Fakenet, SystemInformer
- **Infrastructure & DevOps:** Docker, Linux (Ubuntu/Kali), Git, GitHub, VirtualBox, Google Cloud Platform (GCP)
- **Web Development & Databases:** MERN Stack (MongoDB, Express.js, React.js, Node.js), REST APIs, JWT Authentication, Nodemailer, SQL, Secure Coding Practices, Generative AI (LLMs)
- **Certifications:** Cyber Security 101 Learning Path – TryHackMe
Pre Security Learning Path – TryHackMe
Programming 100: Fundamentals – TCM Security
Supervised Machine Learning: Regression and Classification – DeepLearning.AI
Unsupervised Learning, Recommenders, Reinforcement Learning – DeepLearning.AI
Windows Forensics with Belkasoft
Beyond the Basics: Mastering Advanced Digital Forensics Techniques

Education

• FAST National University of Computer and Emerging Sciences

August 2023 – Present (6th Semester)

Bachelor of Science in Cyber Security

Islamabad, Pakistan

Honors and Activities

- **Honor:** Winner of Programming Competition – University of Mianwali, Jan 2025
- **Activity:** Member, FAST CS Society Operations Team – Sep 2024 – Present

- **Leadership:** House Captain, College (2021 – 2023) – Led a team of 100+ students, developed leadership skill, teamwork, and event management skills.
- **Languages:** English, Urdu
- **Interests:** Competitive Programming, Basketball, E-Games